(i.e., trial and error), because you don’t have a good model of that person’s house.

The trial and error strategy generally falls under the umbrella of *Monte Carlo methods*. A Monte Carlo method is essentially a random sampling from the environment. In many real-world problems, we have at least some knowledge of how the environment works, so we end up employing a mixed strategy of some amount of trial and error and some amount of exploiting what we already know about the environment to directly solve the easy sub-objectives.

A silly example of a mixed strategy would be if you were blindfolded, placed in an unknown location in your house, and told to find the bathroom by throwing pebbles and listening for the noise. You might start by decomposing the high-level goal (find the bathroom) into a more accessible sub-goal: figure out which room you’re currently in. To solve this sub-goal, you might throw a few pebbles in random directions and assess the size of the room, which might give you enough information to infer which room you’re in—say the bedroom. Then you’d need to pivot to another sub-goal: navigating to the door so you can enter the hallway. You’d then start throwing pebbles again, but since you remember the results of your last random pebble throwing, you could target your throwing to areas of less certainty. Iterating over this process, you might eventually find your bathroom. In this case, you would be applying both the goal decomposition of dynamic programming and the random sampling of Monte Carlo methods.

1.4. The reinforcement learning framework

Richard Bellman introduced dynamic programming as a general method of solving certain kinds of control or decision problems, but it occupies an extreme end of the RL continuum. Arguably, Bellman’s more important contribution was helping develop the standard framework for RL problems. The RL framework is essentially the core set of terms and concepts that every RL problem can be phrased in. This not only provides a standardized language for communicating with other engineers and researchers, it also